Reproducibility Project

Kristen Forte, Sebastian Segura, Tanviben Goswami, Keerthana Lingala

Introduction

Paper Summary

We selected a paper, “Introduction of CNN Keras 0.997 Top 6%” by Yassine Ghouzam, which presents a deep learning approach for image classification using Convolutional Neural Networks implemented in Keras. The paper is more focused on training the CNN model with the Kaggle Digit Recognizer dataset, which is based on the MNIST Handwritten Dig dataset. The author seems to use a structured approach when processing, designing the model architecture, and tuning hyperparameters to achieve a high classification accuracy of 0.997. The methodology also seems to include data augmentation, dropout regularization, and the Adam optimizer to prevent overfitting. Despite the high performance, the paper lacks certain explicit details, like specifics of the computational environment and various processing steps. These missing details create potential reproducibility challenges, which we will investigate through this project.

Reproduction Goals

The study will try to replicate Yassine Ghouzam’s CNN model by using indirect replication methods in an attempt to study both methods and see their differences with the original paper and document the discrepancies shown while attempting it while just comparing the results from both.

Methodology

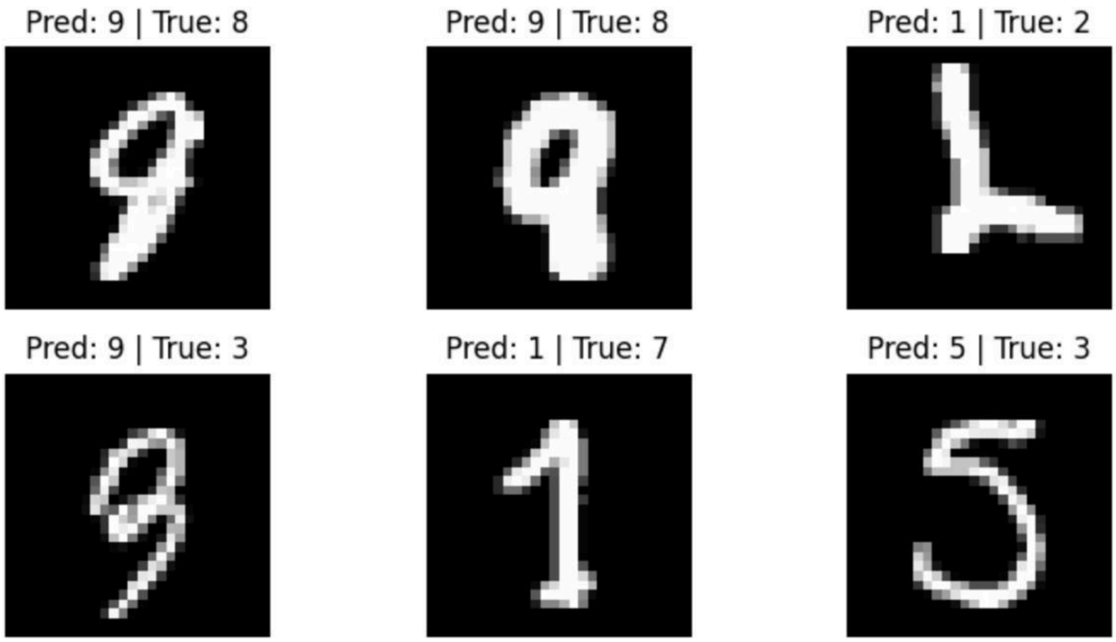
Reproduction Approach

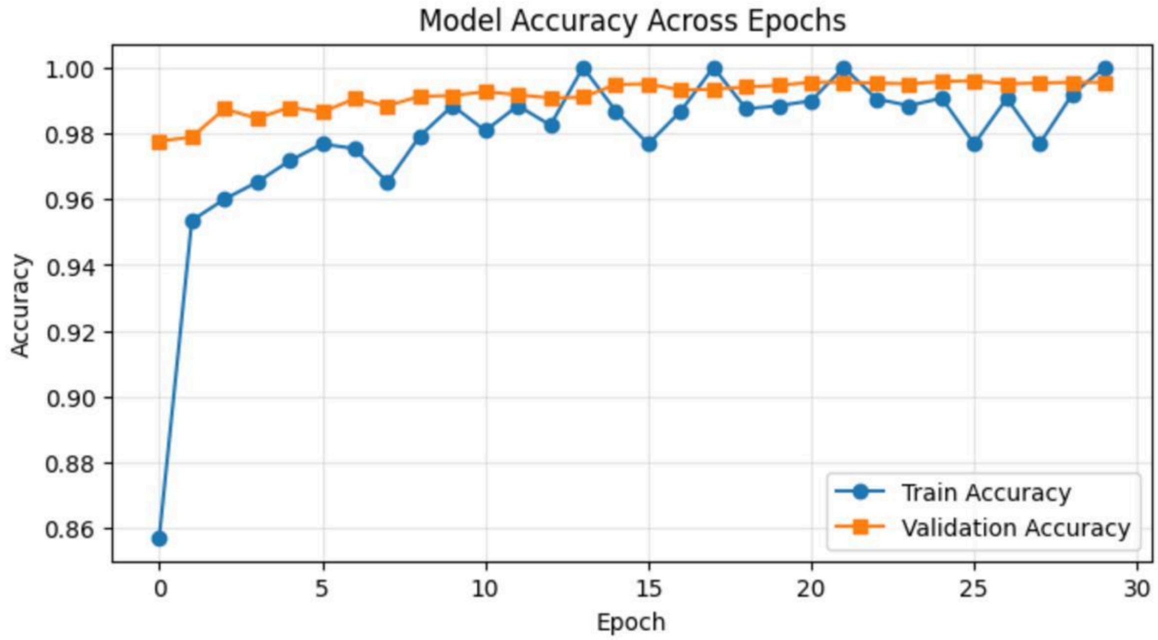
In the indirect reproduction approach, we followed the original CNN notebook closely, but we did not copy every step exactly in the same way. The purpose of this part was to understand how small differences in implementation can affect the reproducibility of the model. Instead of focusing only on exact replication, this approach helps show whether a model still performs well when there are slight variations in execution.

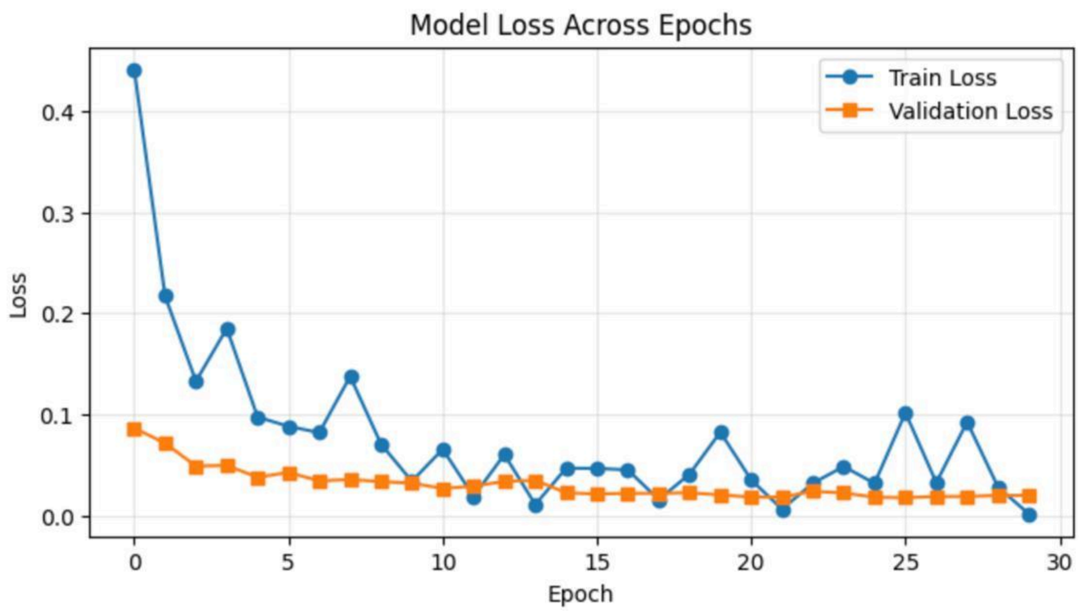
Reproduction Environment and Controlled Variations Introduced

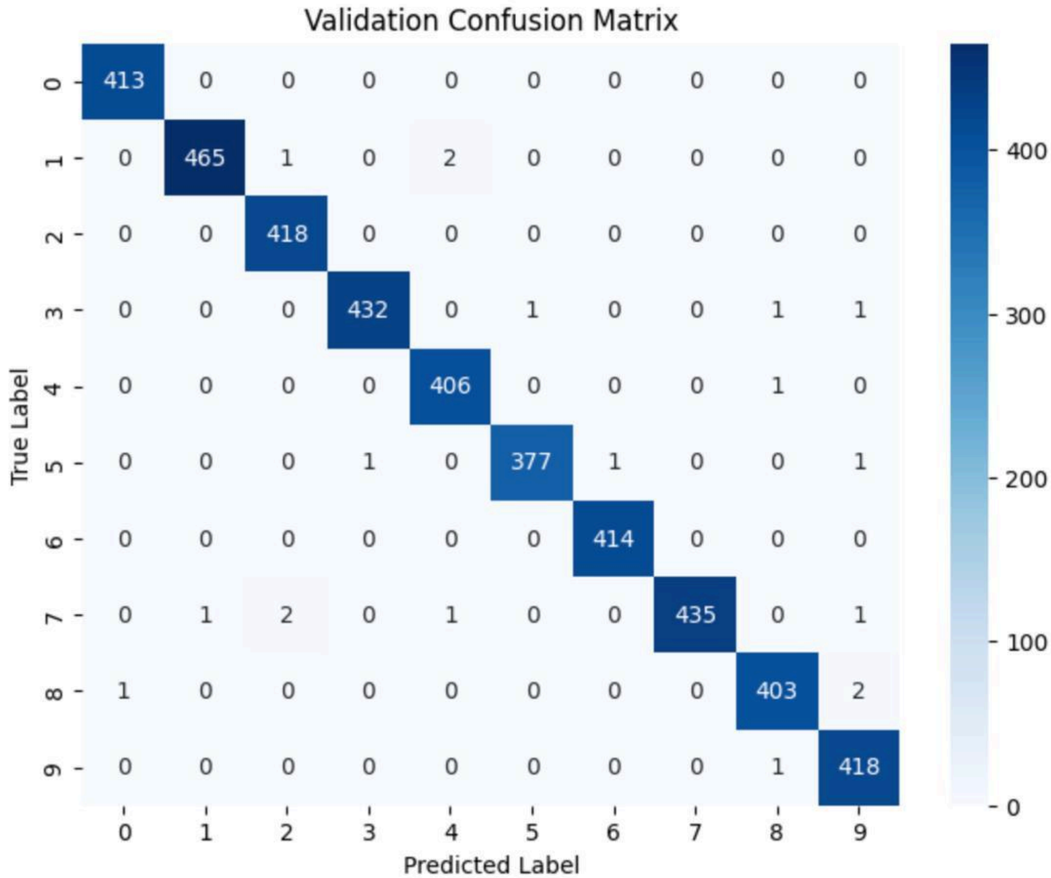
The model was implemented using Python and Jupyter Notebook, following the same general CNN-based method as the original Kaggle notebook. The same digit recognition dataset was used, and the overall training process remained similar to the original work. However, there were slight differences in the implementation compared to the original notebook. These included small variations in execution, settings, or parameter choices during the coding process. Although these changes were minor, they are important because even small differences can influence the final model performance in deep learning experiments.

Reproduction Data Processing Steps and observed effects of controlled variations on the model accuracy




Model Accuracy Across Epochs
Accuracy
Epoch
Train Accuracy
Validation Accuracy
1.00
0.98
0.96
0.94
0.92
0.90
0.88
0.86
0
5
10
15
20
25
30


Model Loss Across Epochs
Loss
Epoch
Train Loss
Validation Loss
0.4
0.3
0.2
0.1
0.0
0
5
10
15
20
25
30



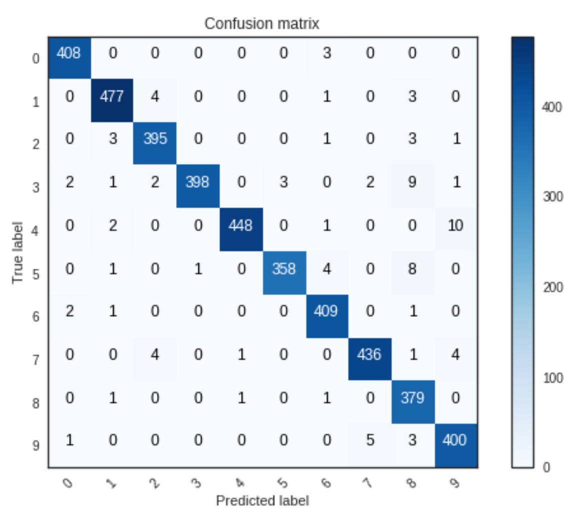
The data processing steps were similar to the original notebook, where the dataset was prepared, reshaped, and used to train the CNN model. After training the model, the final validation accuracy achieved was approximately **99.55%**, which is very close to the original reported accuracy of around **99.7%**.

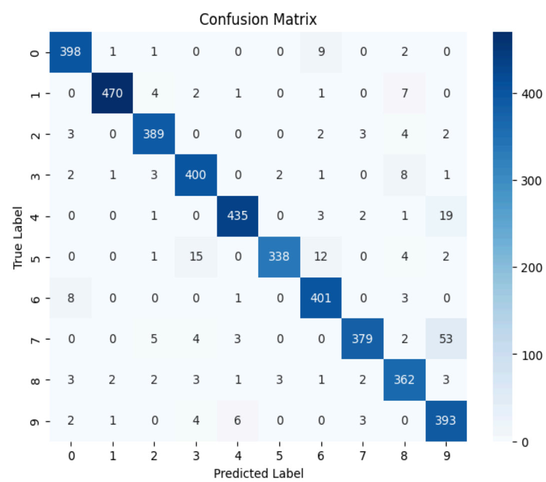
Figure 1 shows sample predictions made by the model, including both correct classifications and a few misclassifications. This indicates that while the model performs very well, it is not perfectly accurate. Figures 2 and 3 present the training and validation accuracy and loss across epochs, showing that the model learned effectively and maintained stable performance during training. Figure 4 displays the confusion matrix, which shows that most digits were correctly classified with only a few errors across different classes.

Even though the model achieved high accuracy, the slight difference compared to the original result highlights that small variations in implementation can affect performance. This demonstrates that reproducibility in machine learning can be challenging, as even minor changes in execution or settings may lead to slightly different outcomes.

Reproduction vs Original

The direct replication aimed to reproduce the original CNN model as closely as possible, following its original work. Despite clearly having issues with outdated software, the results seemed to still be consistent with the original. There were key differences observed; the replication required a different number of epochs to achieve a similar result to the original paper, which only needed 1. Both methods of implementation used batch sizes of 86 to ensure an equal amount in both training batch sizes. Despite some of the changes identified in the model, the performance remained stable while being able to handle the minor software changes.





Figures: The confusion matrix on the left is from the original, and the one on the right is from the exact replication.

Ethics Discussion

Reproducibility Challenges

Overall, challenges were minimal with reproducing this paper, thanks in major part to the availability of the data and code used.

Resource Considerations

What's not often considered when trying to reproduce a paper's results is the difference between the resources you have available and the resources the paper's original authors used. Often, the scientists who wrote the paper will have access to special and more advanced hardware/software that they used during their research. This makes it especially hard for someone to reproduce their results when all they have to work with is their own, less advanced, hardware/software.

Scientific Integrity Implications
Another key point in the ethics of reproducibility is how the ability to reproduce results described by papers directly correlates to that paper's scientific integrity. Results that can't be reproduced are useless in the general scientific community, as there's no way for others to independently verify if the obtained results are true and/or representative of real life.

Recommendations

Best Practices
Good and thorough documentation is the easiest way to ensure your paper is reproducible. Documentation should include:

- The original data
  - Info about how it was obtained
- All processing done
  - What/how features were added
  - How data was train/test split
  - How data was standardized
  - Any other processing steps taken
- All models used
  - What random seeds were used
  - What features were used
  - All parameters used to tune the models
- Environment
  - Programming language
  - Version of software used
  - Any specialized hardware used

By documenting this information, you can greatly improve your paper's reproducibility, and therefore its impact, to the wider scientific community.